

DARIOUS BROWN

Artificial Intelligence & Machine Learning

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PROFILE

AI & Machine Learning Engineer with doctoral-level training in Artificial Intelligence, Machine Learning, and Business Administration, specializing in the design and deployment of scalable, production-grade machine learning systems. Proven ability to build end-to-end pipelines, develop deep learning models, and implement explainable AI solutions that drive measurable business and operational outcomes. Experienced in Python-based development, real-time applications, and system-level AI integration.

EDUCATION

Walsh College
2025-Current

PhD in Artificial Intelligence & ML Specialization

Walsh College
2025-Current

Doctorate of Business Administration

- Focused in Artificial Intelligence

Bellevue University
2025

Masters in Data Science

Saint Joseph's
University
2023

Bachelors in Biomedical Sciences

WORK EXPERIENCE

Invance
2024-Current

Aseptic Technician 2

- Operate within ISO-5/ISO-7 cleanroom environments supporting GMP-compliant autologous TIL cell therapy manufacturing, ensuring full traceability, compliance, and data integrity across high-risk production workflows
- Perform environmental monitoring (viable and non-viable sampling) and conduct trend analysis to identify contamination patterns, supporting proactive risk mitigation and quality optimization
- Designed and implemented structured data tracking frameworks to capture process variability, enabling downstream predictive analytics and system-level visibility into manufacturing performance
- Translate complex manufacturing and environmental datasets into actionable insights, improving operational decision-making and identifying inefficiencies across production cycles
- Collaborate cross-functionally with Quality, MSAT, and Supply Chain teams to align data-driven insights with compliance requirements and operational improvements
- Architecting an AI-driven waste tracking system (dissertation-aligned), incorporating anomaly detection, forecasting models, and decision-support logic within GMP and regulatory constraints
- Bridging manufacturing operations with AI system design, positioning production data for future machine learning integration and predictive optimization

Aseptic Technician

- Executed GMP-aligned molecular biology workflows (PCR, DNA cloning, electrophoresis) within regulated laboratory environments
- Improved operational efficiency by ~15% through process optimization and workflow standardization, reducing variability and increasing throughput
- Structured and curated experimental datasets to enhance reproducibility, traceability, and downstream analytical usability
- Applied data-driven thinking to laboratory operations, aligning experimental execution with quality metrics and reporting requirements
- Maintained strict adherence to SOPs, ensuring regulatory compliance, contamination control, and data integrity across all processes

TECHINICAL SKILLS

- Languages: Python, R, SQL
- Frameworks: Scikit-learn, PyTorch, Jupyter Notebook
- Tools: Streamlit, MLflow, Git,
- ML Techniques: Random Forest, Logistic Regression, LSTM, CNNs, Clustering, Deep Learning, CNN, RNN, LSTM
- Analytics: Time-Series Forecasting, Statistical Modeling, Hypothesis Testing, Machine Learning
- Explainability: SHAP, Feature Importance, Model Diagnostics
- Domain: Biotech Manufacturing (GMP), Healthcare Analytics, Financial Risk Modeling, Waste tracking, Analytical Decision Making

AI PROJECTS & SYSTEMS

- Loan Default Prediction System (Deployed)
 - Developed supervised machine learning models (Random Forest, Logistic Regression) for credit risk prediction
 - Achieved strong predictive performance using ROC-AUC and classification metrics
 - Integrated SHAP explainability to enhance model transparency
 - Deployed Streamlit-based application supporting real-time and batch inference
- Material Waste Tracker (Dissertation Project)
 - Designed AI-ready data architecture for GMP-regulated environments
 - Built Streamlit application with anomaly detection, forecasting, and reporting capabilities
 - Developed predictive framework targeting 10-25% waste reduction
 - Implemented audit-compliant data logging and de-identification protocols
- Bitcoin Price Forecasting (LSTM)
 - Built LSTM deep learning model for time-series forecasting
 - Conducted EDA, PMF/CDF analysis, and regression modeling
 - Identified key drivers including trading volume and market capitalization
- Generative AI Hallucination Detection
 - Fine-tuned GPT-based models to detect hallucinations using confidence scoring
 - Improved reliability and evaluation of generative AI systems